

Amshavarthana Sreesivanathasarma

Software Engineering Undergraduate | Aspiring Data Science & Machine Learning Intern

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🌐 LinkedIn | 🐙 GitHub | 📝 Medium | 🌐 Portfolio

Professional Summary

Software Engineering undergraduate at the University of Kelaniya (GPA 3.71/4.00) with a strong interest in Data Science, Machine Learning, and AI-powered applications. Experienced in Python, SQL, data analysis, and building end-to-end ML solutions using scikit-learn, XGBoost, Flask, and Streamlit. Hands-on project experience spanning predictive modeling, feature engineering, NLP, and LLM-integrated applications, with a focus on rigorous evaluation and deployment.

Education

- **University of Kelaniya** 2024 – Present
BSc (Hons) in Software Engineering **Current GPA: 3.71 / 4.00**
Specialized in Data Science, Net-Centric Application Development, and Business Engineering Applications.
- **J/Arunodaya College** 2020 – 2023
G.C.E. Advanced Level Examination (2023) Combined Mathematics, Chemistry, Physics **Z-Score: 1.5368**

Projects

- **EduMind AI** *Individual* 🌐 Live Demo 🐙 Source Code
Python, Streamlit, Groq LLaMA, scikit-learn, XGBoost, SQLite, Plotly
 - Built a **trilingual (English, Tamil, Sinhala)** AI study companion for Sri Lankan A/L and university students, powered by **Groq LLaMA**.
 - Trained **Random Forest** and **XGBoost** classifiers on **6,389 student records** across 5 datasets, achieving **93.85% accuracy** for academic risk prediction and **85.31%** for dropout detection.
 - Integrated trained models into a live **Streamlit** app for real-time student risk assessment, visualized with an interactive **Plotly** heatmap backed by SQLite.
- **Password Strength Classifier** *Individual* 🐙 Source Code
Python, Scikit-learn, Pandas, NumPy, SQLite
 - Engineered a **104-dimensional feature set** (99 character-level TF-IDF features + 5 structural features) from approximately **100,000 leaked passwords** to classify strength via multinomial **Logistic Regression**.
 - Achieved **80% accuracy** and **0.78 weighted F1-score** on a 20,000-sample test set.
 - Identified class imbalance as the key driver of lower recall on minority classes, informing future work on resampling and n-gram features.
- **Flight Fare Prediction System** *Individual* 🐙 Source Code
Python, Pandas, NumPy, Scikit-learn, Flask, HTML
 - Performed **EDA** and **feature engineering** on historical flight data (airline, route, duration, stops) to prepare inputs for ML modelling.
 - Developed a **Random Forest Regressor** achieving $R^2 = 0.81$ and **MAPE = 13.3%** on ticket fare prediction.
 - Deployed the model via a **Flask** web interface for real-time fare estimation.

Technical Skills

- **Languages:** Python, SQL, Java, JavaScript
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, XGBoost, TF-IDF, MLP Neural Networks, Feature Engineering, Model Evaluation
- **AI / LLM Integration:** Groq LLaMA, OpenAI Whisper, Tavily API
- **Frameworks:** Flask, Streamlit, Spring Boot
- **Databases:** MySQL, SQLite
- **Visualization:** Matplotlib, Seaborn, Plotly, Power BI

Certifications

- **Data Science Real World Projects in Python** – Udemy 🔗 View Certificate
- **The Complete Python Masterclass** – Udemy 🔗 View Certificate
- **Statistics Foundations** – LinkedIn Learning 🔗 View Certificate
- **Machine Learning with Python Professional Certificate** – LinkedIn Learning (Anaconda) 🔗 View Certificate

Soft Skills & Languages

Soft Skills: Team Collaboration, Time Management, Critical Thinking, Problem Solving, Adaptability & Quick Learning

Languages: Tamil (Native), English (Fluent), Sinhala (Basic)

References

- **Prof. (Mrs.) I. U. Hewapathirana**, Senior Lecturer, Faculty of Science, University of Kelaniya
✉ ihewapathirana@kln.ac.lk
- **Dr. Nalin Warnajith**, Senior Lecturer, Faculty of Science, University of Kelaniya
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